

Human activity recognition using multimodal fusion strategies

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Abstract

Human Activity Recognition (HAR) systems often rely on heterogeneous sensor data and feature representations, where a single model may not perform optimally across all activities. Model fusion techniques, such as Mixture of Experts (MoE) and stacking, address this limitation by combining multiple specialized models to exploit their complementary strengths. In MoE, a gating mechanism dynamically selects or weights expert predictions, while stacking learns a meta-model over expert outputs. These approaches improve robustness, generalization, and adaptability to inter-subject variability. Experimental results show that fusion models consistently outperform individual classifiers, making them well-suited for complex and real-world HAR scenario

1 Introduction

Over the last decade, wearable inertial sensors have become essential tools for large-scale population research and digital health monitoring. Their widespread use across national and international cohorts has enabled the objective characterization of human behavior at an unprecedented scale. Yet, despite notable progress in activity recognition and behavioral phenotyping, most cohort studies continue to rely on data from a *single* sensor location—typically the wrist or thigh—thus limiting the potential to exploit the complementary information that arises from multisite monitoring.

Wrist-worn devices offer high compliance and are well suited for capturing upper-limb dynamics, but they remain suboptimal for assessing posture allocation or sedentary behavior. Conversely, thigh-mounted sensors provide accurate estimates of sitting, standing, and postural transitions, but they lack sensitivity to fine-grained arm movements that are frequent in free-living conditions. This natural complementarity suggests that combining wrist and thigh signals could yield more robust and generalizable activity-recognition models. However, systematic investigations into how to *integrate* heterogeneous IMU streams from multiple body locations—and how different integration strategies impact downstream activity-classification performance—remain scarce.

Recent cohort initiatives that include both wrist and thigh IMUs, such as emerging next-generation precision-health studies, highlight the growing need for principled multisensor fusion methodologies. In this context, the community still lacks comparative evidence on whether simple feature-level fusion, probabilistic ensembling, or advanced mixture-of-experts (MoE) strategies provide the most reliable performance gains. Moreover, little is known about how these integration approaches behave under realistic constraints typical of population research, such as variable sensor quality, imperfect synchronization, or missing modality scenarios.

The present study addresses these gaps by conducting a comprehensive evaluation of multimodal fusion techniques for combining wrist- and thigh-mounted IMUs in physical-activity classification tasks. Using data from the WearablePerMed project, we investigate a spectrum of integration strategies ranging from classical stacking and decision-level ensembling to modern MoE architectures specifically tailored for heterogeneous sensor modalities. Our analysis quantifies the contribution of each fusion approach at multiple levels of activity granularity and examines the extent to which multimodal integration provides consistent advantages over single-sensor baselines.

By systematically comparing these strategies under matched experimental conditions, this work provides new methodological insights into the design of multisite wearable-sensor systems and estab-

lishes evidence-based guidelines for future national cohort deployments that seek to exploit the joint potential of wrist–thigh sensing configurations.

2 Materials and Methods

2.1 Dataset

The employed dataset comprises data from 60 participants. Linear acceleration and angular velocity data from the thigh, wrist, and hip were acquired using inertial measurement units (IMUs) from the MATRIX family, which integrate a triaxial accelerometer and a triaxial gyroscope. These sensors operate with a full-scale range of ± 16 g and a resolution of 16 bits. Previous studies have reported that the accelerometer noise level is below 3 mg, while the gyro-scope exhibits a higher noise level of approximately $0.12^\circ/\text{s}$ [1].

2.2 Data Preprocessing

3 Results

F1-score metrics (classification performance) are reported for the test dataset in Table 1 detailing the fusion model (Individual Model, Concatenated, Stacking Random Forest, Stacking Autoencoders, Mixture of Experts Random Forest, Mixture of Experts Autoencoder and Mixture of Experts Variational Autoencoder), and the granularity of the classification (4 activity classes, 8 daily activity classes and 15 daily activity classes). Mean values, along with the associated standard deviation of test classification performance from 30 independent runs are reported intensity levels.

Figure 1 summarizes the F1-score distributions obtained for the different model configurations across three labeling granularities (4, 8, and 15 classes). Each boxplot represents the variability across cross-validation folds (or repetitions), enabling comparison between single-sensor baselines, early fusion, and late fusion approaches.

3.1 Performance with 4 Classes

For the simplest labeling scheme (4 classes), fusion strategies consistently outperform individual sensor models. The single-sensor baselines show median F1-scores around 79% (thigh, PI) and 76% (wrist, M). Early fusion through feature concatenation improves performance to approximately 86%–88%, confirming that combining complementary sensor streams increases discriminative power.

Among late fusion approaches, stacking with Random Forest base learners (stacking_rf) achieves the highest performance (median $\approx 90\%$ – 91%), followed by the mixture of experts with Random Forest (moe_rf) and the concatenated model. In contrast, stacking_ae and moe_ae exhibit higher variance and lower medians, suggesting that autoencoder-based latent representations are less effective for this simpler classification problem. The moe_vae configuration yields intermediate performance but remains below Random Forest–based fusion methods.

3.2 Performance with 8 Classes

Increasing the number of classes to 8 slightly reduces overall performance but preserves the same ranking across models. Individual sensors degrade to approximately 90% (PI) and 72%–74% (M). Early fusion again provides a substantial improvement (92%–94% median F1).

The best-performing models remain stacking_rf and moe_rf, both achieving median F1-scores around 93%–95% with relatively low dispersion. Autoencoder-based stacking and MoE approaches again present higher variability and lower central tendency, particularly moe_ae, which shows notable outliers and instability. The moe_vae configuration provides a stable but slightly lower performance ($\sim 80\%$ – 82%).

3.3 Performance with 15 Classes

The most challenging scenario (15 classes) leads to a general performance decrease across all models, as expected due to increased class granularity. Individual models drop to medians of approximately 60% (PI) and 69% (M). Early fusion remains beneficial, raising performance to about 79%–82%.

Late fusion models continue to provide the best results. `stacking_rf` remains the top performer with median F1 around 83%–85%, closely followed by `moe_rf` (~77%–80%). The gap between Random Forest–based fusion and autoencoder-based approaches becomes more pronounced in this setting. Both `stacking_ae` and `moe_ae` show lower medians (~65%–70%) and increased variance, while `moe_vae` achieves moderate performance (~68%–71%).

3.4 Overall Trends

Across all labeling schemes, several consistent trends emerge:

1. **Fusion improves performance:** both early fusion (feature concatenation) and late fusion (stacking and MoE) outperform individual sensor models, supporting the hypothesis that multi-sensor information is complementary.
2. **Late fusion is superior to early fusion:** stacking and mixture-of-experts approaches generally achieve higher median F1-scores than simple concatenation.
3. **Random Forest–based fusion is the most effective:** models using Random Forest as base learners or experts consistently yield the highest performance and stability.
4. **Representation learning approaches show higher variance:** autoencoder and VAE-based models underperform compared to tree-based methods, particularly in more complex classification scenarios.
5. **Performance decreases with class granularity, but fusion methods remain advantageous:** although performance drops as the number of classes increases, fusion methods maintain a consistent advantage even in the most challenging 15-class setting.

These findings align with previous work demonstrating the benefits of multimodal fusion and ensemble learning for activity recognition and sensor-based classification tasks (e.g., Dietterich, 2000; Rokach, 2010; Atrey et al., 2010).

Table 1: F1-score (mean $\pm$ standard deviation) for each fusion strategy and granularity level.

Fusion Strategy	Granularity	F1 Score (%)
individual_pi	4	79.29 $\pm$ 2.25
individual_pi	8	89.94 $\pm$ 2.14
individual_pi	15	59.61 $\pm$ 2.70
individual_m	4	76.85 $\pm$ 1.85
individual_m	8	72.66 $\pm$ 2.33
individual_m	15	69.70 $\pm$ 2.67
concatenated	4	86.35 $\pm$ 1.74
concatenated	8	92.02 $\pm$ 2.16
concatenated	15	79.76 $\pm$ 3.27
stacking_rf	4	89.81 $\pm$ 1.51
stacking_rf	8	93.07 $\pm$ 2.70
stacking_rf	15	82.50 $\pm$ 2.48
stacking_ae	4	70.48 $\pm$ 8.23
stacking_ae	8	72.44 $\pm$ 8.27
stacking_ae	15	68.56 $\pm$ 6.13
moe_rf	4	84.00 $\pm$ 1.32
moe_rf	8	92.10 $\pm$ 1.50
moe_rf	15	77.11 $\pm$ 2.94
moe_ae	4	74.33 $\pm$ 3.44
moe_ae	8	80.66 $\pm$ 4.32
moe_ae	15	63.43 $\pm$ 5.68
moe_vae	4	69.31 $\pm$ 1.29
moe_vae	8	80.84 $\pm$ 2.02
moe_vae	15	67.98 $\pm$ 3.03

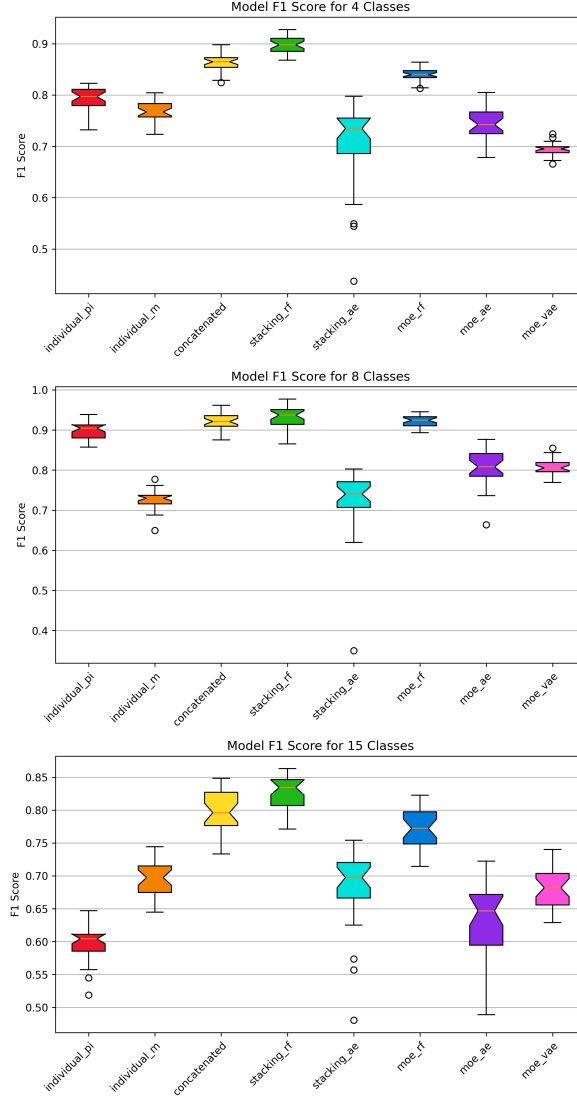


Figure 1: Model Fusion comparison

4 Discussion

This study provides a systematic comparison of multiple multimodal fusion strategies for integrating wrist- and thigh-mounted IMU data in physical-activity classification. Across all labeling granularities (4, 8, and 15 classes), the results consistently demonstrate that multimodal fusion improves performance over single-sensor baselines, confirming the complementary nature of upper- and lower-limb motion signals in free-living conditions.

Among the evaluated approaches, late fusion strategies—particularly `stacking_rf` and `moe_rf`—consistently achieved the highest F1-scores and the lowest dispersion across folds. This finding suggests that ensemble-based decision-level integration is more effective than simple feature concatenation when combining heterogeneous sensor modalities. In particular, the superior performance of Random Forest-based fusion methods indicates that tree-based learners are well suited to capturing non-linear interactions between modality-specific representations while remaining robust to noise and variability in sensor signals.

The advantage of late fusion over early fusion becomes more evident as the classification problem becomes more complex. While feature concatenation provides clear gains over single-sensor models, its performance saturates more quickly as the number of classes increases. In contrast, stacking and mixture-of-experts (MoE) models maintain stronger performance under higher class granularity, sug-

gesting better scalability and improved capacity to model modality-specific uncertainty. This behavior is especially relevant for population-scale studies, where activity taxonomies often involve fine-grained behavioral categories.

In contrast, representation learning approaches based on autoencoders and variational autoencoders show higher variability and consistently lower median performance compared to Random Forest-based methods. Although these approaches offer a principled way to learn latent multimodal representations, their sensitivity to training conditions and limited performance gains in this context suggest that they may require larger datasets or more sophisticated regularization strategies to outperform classical ensemble techniques. Additionally, the increased variance observed in `stacking_ae` and `moe_ae` indicates reduced stability, which may limit their applicability in real-world deployments where robustness is critical.

From a methodological perspective, these results highlight that the choice of fusion strategy should be driven not only by predictive performance but also by robustness, interpretability, and computational cost. Random Forest-based fusion methods provide a favorable balance between these factors, offering strong performance, relatively low variance, and straightforward implementation. This makes them particularly attractive for large cohort studies and digital health applications, where scalability and reproducibility are essential.

Finally, the consistent improvements observed across all class granularities reinforce the value of multisite sensing configurations in wearable-based activity recognition. By leveraging complementary information from wrist and thigh sensors, multimodal fusion approaches can mitigate the limitations inherent to single-sensor setups and provide more reliable behavioral characterization in free-living environments. These findings support the growing interest in multisensor deployments within next-generation population cohorts and precision health initiatives.

5 Limitations of the study

Signal patterns were visually inspected to identify segmentation markers, which may have introduced some inaccuracies and resulted in a small number of mislabelled windows within the dataset. Additionally, we observed that certain activities for a few subjects exhibited brief periods of inactivity, possibly due to interruptions during the task, that could not be removed, occasionally leading to incorrectly labelled segments. However, the impact of these issues on overall classification performance is expected to be minor.

6 Conclusion

This work presented a comprehensive evaluation of multimodal fusion strategies for combining wrist- and thigh-mounted IMU data in physical-activity classification tasks. Through systematic experiments across multiple labeling granularities, we demonstrated that multimodal integration consistently outperforms single-sensor approaches, confirming the complementary value of multisite wearable sensing.

Among the evaluated methods, Random Forest-based late fusion models—in particular `stacking_rf` and `moe_rf`—achieved the best overall performance and stability, outperforming both early fusion (feature concatenation) and representation learning-based approaches. These results indicate that ensemble-based decision-level fusion provides a robust and scalable solution for integrating heterogeneous sensor modalities in real-world settings.

The study also revealed that the benefits of multimodal fusion become more pronounced as task complexity increases, highlighting the importance of advanced fusion strategies for fine-grained activity recognition. In contrast, autoencoder- and VAE-based methods showed higher variability and lower performance, suggesting that further methodological developments are needed before such approaches can fully exploit multimodal wearable data in population-scale scenarios.

Overall, our findings provide practical guidelines for the design of multisensor wearable systems and demonstrate that combining wrist and thigh IMUs using robust ensemble fusion techniques yields consistent and meaningful performance gains. These insights contribute to the development of next-generation digital health platforms and large-scale cohort studies aiming to achieve more accurate and comprehensive characterization of human behavior.

Future work should explore the integration of additional sensing modalities, investigate adaptive fusion mechanisms under missing-data conditions, and evaluate the generalization of these models in fully free-living, longitudinal cohort settings.

References

- [1] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [2] Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
- [3] Kuhn, M. and Johnson, K. (2019). *Feature Engineering and Selection: A Practical Approach for Predictive Models*. CRC Press.